1. COMPUTING WITH REAL NUMBERS

Floating-point system

Normalized binary FPN: $\mathrm{fl}(x)=\pm(1.d_1d_2\ldots d_p)_2\,2^e$, $d_j\in\{0,1\}$, $e_{min}\le e\le e_{max}$. IEEE double: sign 1 bit, p=52 fraction bits, 11 exponent bits; normal $e\in[-1022,1023]$.
$\varepsilon=2^{-52}\approx2.22\times10^{-16}$ (gap from 1 to next FPN);
$\varepsilon_i=2^m\varepsilon\ (2^m\le i<2^{m+1})$, $\quad x_{\max}=(2-\varepsilon)2^{1023}$, $\quad x_{\min}^{\text{normal}}=2^{-1022}$. Below: subnormal, then underflow to 0; overflow→±∞; undefined→NaN. ***FPN arithmetic is not associative; avoid exact equality tests.***

Errors, accuracy, stability

For approximation $\tilde{x}$ to x: $e_a=|\tilde{x}-x|$, $\quad e_r=\frac{|\tilde{x}-x|}{|x|}$ (***x≠0***).

Significant decimal figures: largest p satisfying $e_r<\frac{1}{2}10^{-p}$.

Storage rounding error $\le\varepsilon|x|$.
Catastrophic cancellation: subtracting computed nonzero $\tilde{x},\tilde{y}$ when $x\approx y$ can greatly amplify relative error.
Quadratic $ax^2+bx+c=0$:
$q=-\frac{1}{2}\left(b+\mathrm{sign}(b)\sqrt{b^2-4ac}\right)$, $\quad x_1=\frac{q}{a}$, $\quad x_2=\frac{c}{q}$. ***Require a≠0, real-root form $b^2-4ac\ge0$, q≠0 for c/q.***

Cost and storage

1 bit∈{0,1}; 1 byte=8 bits; KiB=2^{10} bytes, MiB=2^{20}, GiB=2^{30}. Signed b-bit integer range $[-2^{b-1},2^{b-1}-1]$. 1 flop = one FPN +, −, ×, /.
Speed=(cycles/s)(cores)(flops/cycle); time=flops/speed.
$\mathrm{memory}(n\times n)=8n^2$ bytes, $\quad \mathrm{flops}(AB)=n^2(2n-1)\approx2n^3$.

2. NORMS, EQUALITY, SENSITIVITY

Vector norms and error

For $x\in\mathbb{R}^n$, p≥1: $\|x\|_p=\left(\sum_{i=1}^n|x_i|^p\right)^{1/p}$;

$\|x\|_1=\sum_i|x_i|$, $\quad \|x\|_2=\sqrt{x^Tx}$, $\quad \|x\|_\infty=\max_i|x_i|$. Norm axioms: positivity/definiteness, triangle inequality, absolute homogeneity. $x=y\iff\|x-y\|=0$, $\quad \|x-y\|\le\tau(\varepsilon,n,\|x\|,\|y\|)$.

$e_a=\|\tilde{x}-x\|$, $\quad e_r=\frac{\|\tilde{x}-x\|}{\|x\|}$ (***x≠0***);

$\frac{\|\tilde{x}-x\|_\infty}{\|x\|_\infty}<\frac{1}{2}10^{-k}$, $\quad k=\lfloor-\log_{10}(2e_{r,\infty})\rfloor$.

Matrix norms

Subordinate norm: $\|A\|_p=\max_{x\ne0}\frac{\|Ax\|_p}{\|x\|_p}$;

$\|Ax\|_p\le\|A\|_p\|x\|_p$, $\quad \|AB\|_p\le\|A\|_p\|B\|_p$.

$\|A\|_1=\max_j\sum_i|a_{ij}|$, $\quad \|A\|_\infty=\max_i\sum_j|a_{ij}|$, $\quad \|A\|_2=\sqrt{\lambda_{\max}(A^TA)}=\sigma_1$;

Frobenius $\|A\|_F=\sqrt{\sum_{ij}|a_{ij}|^2}$ (***not generally $\|A\|_2$***).

Conditioning of Ax=b

For nonsingular A: $\kappa(A)=\|A\|\,\|A^{-1}\|\ge1$; rcond=1/κ.
$\kappa_2(A)=\frac{\max_i|\lambda_i|}{\min_i|\lambda_i|}\ (A=A^T)$, $\quad \kappa_2(A)=\frac{\sigma_1}{\sigma_n}\ (\mathrm{rank}(A)=n)$.
$(A+\delta A)(x+\delta x)=b+\delta b$, $\quad \|(A+\delta A)^{-1}\|\le\alpha\|A^{-1}\|$; A+δA nonsingular, α>1. $\rho_A=\frac{\|\delta A\|}{\|A\|}$, $\quad \rho_b=\frac{\|\delta b\|}{\|b\|}$; $\frac{\|\delta x\|}{\|x\|}\le\alpha\kappa(A)(\rho_A+\rho_b)$.
Large κ ⇒ sensitive/ill-conditioned; κ≳1/ε can destroy machine-precision information.

3. MATRIX STRUCTURE & FACTORISATIONS

Special matrices

$C_{ij}=E[(X_i-\mu_i)(X_j-\mu_j)]$, $\quad C_{ii}=\sigma_i^2$; C symmetric PSD.

$K_{ij}=\frac{C_{ij}}{\sigma_i\sigma_j}$, $\quad K=\Sigma^{-1}C\Sigma^{-1}$, $\quad K_{ii}=1$, $|K_{ij}|\le1$ (***$\sigma_i,\sigma_j>0$***).

Upper/lower triangular: $a_{ij}=0$ for i>j / i<j.
$\det A=\prod_i a_{ii}$, $\quad A$ nonsingular $\iff a_{ii}\ne0\ \forall i$. Unit triangular: $a_{ii}=1$.
Triangular solve ≈n^2 flops.
$a_{ij}=0\quad\text{if}\quad i-j>m_\ell\text{ or }j-i>m_u$; total bandwidth $m_\ell+m_u+1$; nnz≤$(1+m_\ell+m_u)n-m_\ell(m_\ell+1)/2-m_u(m_u+1)/2$. Tridiagonal: $m_\ell=m_u=1$. Toeplitz: $a_{ij}=\alpha_{j-i}$.
Symmetric $A^T=A$; skew $A^T=-A$. $A=\frac{A+A^T}{2}+\frac{A-A^T}{2}$. Definiteness by x^TAx: PD >0, PSD ≥0, ND <0, NSD ≤0; for real symmetric A, signs of eigenvalues give type. Strict row diagonal dominance: $|a_{ii}|>\sum_{j\ne i}|a_{ij}|$; symmetric + strict dominance ⇒ PD.
$Q^TQ=I$, $\quad Q^{-1}=Q^T$, $\quad \kappa_2(Q)=1$, $\quad \|Qx\|_2=\|x\|_2$.
Permutation matrices are orthogonal.

LU and Cholesky

LU: A=LU with L unit lower, U upper; when it exists for nonsingular A it is unique. In general partial pivoting gives $PA=LU$; $|\ell_{ij}|\le1$. Factor cost ≈$2n^3/3$; two triangular solves ≈$2n^2$. ***Require nonsingular A for a unique solve; do not form A^{-1}.***
Cholesky: $A=R^TR$, R upper. ***Require real symmetric positive-definite A.***; factor cost $n^3/3$, solves $2n^2$; no row pivoting.

QR

For $A\in\mathbb{R}^{m\times n}$, m≥n, rank(A)=n:
$A=Q\begin{bmatrix}R\\0\end{bmatrix}=YR$, $\quad Q=[Y\ Z]$, $\quad Q^TQ=I$; R nonsingular upper.
Cost ≈$2mn^2-(2/3)n^3$. If rank deficient: $AP=Q\begin{bmatrix}R\\0\end{bmatrix}$, column pivoting, $|r_{11}|\ge\cdots\ge|r_{nn}|$; small diagonal entries indicate numerical rank deficiency.

SVD

$A\in\mathbb{R}^{m\times n}$, m≥n: $A=U\begin{bmatrix}\Sigma\\0\end{bmatrix}V^T$; U,V orthogonal,
$\Sigma=\mathrm{diag}(\sigma_1\ge\cdots\ge\sigma_n\ge0)$. rank(A)=number of positive σ_i;
$A=\sum_{j=1}^r\sigma_ju_jv_j^T$, $\quad Av_i=\sigma_iu_i$, $\quad u_i^TA=\sigma_iv_i^T$. Full rank: $\kappa_2(A)=\sigma_1/\sigma_n$.
Numerical rank tolerance ≈$\max(m,n)\varepsilon\sigma_1$.

4. PARAMETRIC DATA FITTING

Residual model

Data (t_i,y_i), i=1,…,m; parameters $x\in\mathbb{R}^n$, ***m≥n***.
$r_i(x)=\phi(x;t_i)-y_i$, $\quad x^*=\arg\min_x\|r(x)\|_p$. Interpolation ⇔ r=0 (***appropriate only for highly accurate data***).
Linear-in-parameters:
$\phi(x;t)=\sum_jx_j\phi_j(t)$, $\quad A_{ij}=\phi_j(t_i)$, $\quad r=Ax-y$.

Linear least squares (p=2)

Normal equations: $A^TAx^*=A^Ty$; residual orthogonal to col(A). A^TA PSD, and PD iff A has full column rank. ***Avoid forming A^TA when conditioning matters: $\kappa_2(A^TA)=\kappa_2(A)^2$.***
Pivoted QR AP=Q[R;0], Q=[Y Z]: $R(P^Tx^*)=Y^Ty$. SVD $A=U[\Sigma;0]V^T$, U=[Y Z]: $x^*=V\Sigma^{-1}Y^Ty$, $\quad x_{\min}^*=V\Sigma^+Y^Ty$. Use $\Sigma^+=\mathrm{diag}(1/\sigma_1,\ldots,1/\sigma_r,0,\ldots)$ for rank r<n.

Choice of norm / transformation

p=2: suited to independent N(0,σ²) errors with common variance. p=1: robust to outliers,
$\min\sum_iv_i:\quad -v_i\le r_i\le v_i,\quad v_i\ge0$. p=∞: $\min v:\quad -v\le r_i\le v,\quad v\ge0$.
Linear r gives linear programs.
Log-linearisation: products/exponentials/powers may be transformed. ***Log transform requires every logged quantity positive and any shifted base $t-\beta_j>0$.***

5. CUBIC SPLINES & SPARSE MATRICES

Polynomials and interpolation

$p(t)=\sum_{k=0}^na_kt^k=a_0+t(a_1+t(a_2+\cdots+t(a_{n-1}+ta_n)))$; Horner form uses n multiplies+n adds. Weierstrass: for continuous f on closed [a,b], ∀ε>0 ∃ polynomial p with |f−p|<ε on [a,b].
Polynomial interpolation at distinct $t_0,\ldots,t_n$: Vandermonde Ax=y, $A_{ij}=t_i^j$. ***High-degree equally spaced interpolation can be ill-conditioned and oscillatory; use low-degree pieces.***

Cubic spline

Knots $a=t_0<\cdots<t_n=b$. $s(t)=s_i(t)$ on $[t_{i-1},t_i]$, $\deg(s_i)\le3$, and $s\in C^2[a,b]$. Interpolation $s(t_i)=y_i$; at interior knots
$s_i(t_i)=s_{i+1}(t_i)$, $\quad s_i'(t_i)=s_{i+1}'(t_i)$, $\quad s_i''(t_i)=s_{i+1}''(t_i)$. Two endpoint conditions required:
$\text{natural}: s''(a)=s''(b)=0$; $\quad \text{clamped}: s'(a),s'(b)$; $\quad \text{not–a–knot}: s_1'''(t_1)=s_2'''(t_1),\ s_{n-1}'''(t_{n-1})=s_n'''(t_{n-1})$.
Uniform $h=t_i-t_{i-1}$, $d_i=(y_i-y_{i-1})/h$, $M_i=s''(t_i)$. For $t\in[t_{i-1},t_i]$:
$s_i(t)=\frac{M_{i-1}(t_i-t)^3}{6h}+\frac{M_i(t-t_{i-1})^3}{6h}+\left(\frac{y_{i-1}}{h}-\frac{M_{i-1}h}{6}\right)(t_i-t)+\left(\frac{y_i}{h}-\frac{M_ih}{6}\right)(t-t_{i-1})$
Interior system: $\frac{h}{6}M_{i-1}+\frac{2h}{3}M_i+\frac{h}{6}M_{i+1}=d_{i+1}-d_i$, i=1,…,n−1, plus endpoint equations. Natural: $M_0=M_n=0$; coefficient matrix symmetric tridiagonal diagonally dominant PD; Thomas solve O(n).

5. CUBIC SPLINES & SPARSE MATRICES (CONT.)

Sparse structure

Sparsity=#zeros/(mn); density=nnz/(mn). Diagonal density≈1/n; tridiagonal≈3/n. Hidden sparse product:
$(I+BB^T)x=x+B(B^Tx)$, $\quad B\in R^{n\times r}$, $\quad r\ll n$. Store/operate on nonzeros. Fill-in=new nonzeros during factorisation; PAQ=LU can reduce fill-in. For banded A=LU without disruptive pivoting, L/U retain lower/upper bandwidth; SPD $A=R^TR$ gives R same upper bandwidth.

1-D boundary-value finite difference

$-u''=f$ on $(0,1)$, $\quad u(0)=\alpha$, $\quad u(1)=\beta$, $\quad x_j=j\Delta x$, $\quad \Delta x=\frac{1}{n+1}$.
$-\frac{U_{j+1}-2U_j+U_{j-1}}{(\Delta x)^2}=f(x_j)$, j=1,…,n; $U_0=\alpha$, $U_{n+1}=\beta$; error $O((\Delta x)^2)$ under sufficient smoothness. Matrix tridiagonal Toeplitz SPD with diagonal 2 and off-diagonal −1; RHS endpoints add α,β.

6. FOURIER ANALYSIS & FFT

Periodic and trigonometric form

$g(t+T)=g(t)$, $\quad f=\frac{1}{T}$, $\quad \cos(\omega t):T=\frac{2\pi}{\omega}$. Harmonic k: frequency k/T, angular frequency 2πk/T.
$g_n(t)=a_0+\sum_{k=1}^n\left[a_k\cos\left(\frac{2\pi kt}{T}\right)+b_k\sin\left(\frac{2\pi kt}{T}\right)\right]$.
Over any full period (shown on [−T/2,T/2]): $a_0=\frac{1}{T}\int_{-T/2}^{T/2}g(t)\,dt$;
$a_k=\frac{2}{T}\int_{-T/2}^{T/2}g(t)\cos\left(\frac{2\pi kt}{T}\right)dt$; $b_k=\frac{2}{T}\int_{-T/2}^{T/2}g(t)\sin\left(\frac{2\pi kt}{T}\right)dt$. Orthogonality: equal nonzero sine/sine or cosine/cosine integral=T/2; unequal or sine/cosine=0.
If periodic g is piecewise differentiable: Fourier series converges to g at continuity points and to [g(t+)+g(t−)]/2 at jumps. Gibbs oscillations persist near jumps.

Complex Fourier series

$g(t)=\sum_{k\in Z}c_ke^{i2\pi kt/T}$, $\quad c_k=\frac{a_k-ib_k}{2}$, $\quad c_{-k}=\frac{a_k+ib_k}{2}$; $c_k=\frac{1}{T}\int_{-T/2}^{T/2}g(t)e^{-i2\pi kt/T}\,dt$;
$c_0=a_0$. ***Pointwise equality stated at continuity points.***

DFT / inverse / FFT

$\omega=e^{i2\pi/n}$. For j,k=0,…,n−1: $\hat{y}_k=\sum_{j=0}^{n-1}y_je^{-i2\pi jk/n}=\sum_{j=0}^{n-1}y_j\omega^{-jk}$;

$y_j=\frac{1}{n}\sum_{k=0}^{n-1}\hat{y}_ke^{i2\pi jk/n}$. Direct $O(n^2)$; FFT $O(n\log_2n)$.

$y_j\in R:\quad \hat{y}_{n-k}=\overline{\hat{y}_k}$, $\quad \hat{y}_0\in R$, $\quad \hat{y}_{n/2}\in R\ (n\text{ even})$.

Fourier coefficients from equally spaced data

$t_j=j\Delta t$, $\Delta t=T/n$, $y_j\approx g(t_j)$, $p=\lfloor n/2\rfloor$.
$A_0=\frac{\hat{y}_0}{n}$, $\quad A_k=\frac{2\mathrm{Re}(\hat{y}_k)}{n}$, $\quad B_k=-\frac{2\mathrm{Im}(\hat{y}_k)}{n}$.
If n=2p even: k=1,…,p−1 use ordinary form; $A_p=\hat{y}_p/n$, $B_p=0$. If n=2p+1 odd: ordinary form through k=p. Then $y_j=A_0+\sum_k[A_k\cos(2\pi kt_j/T)+B_k\sin(2\pi kt_j/T)]$.

De-trending and spectrum

For $Y_j\approx x_1+x_2t_j+g(t_j)$, g periodic: least-squares fit x_1+x_2t and define $y_j=Y_j-(x_1+x_2t_j)$ before DFT. Power spectrum $p_k=\frac{|\hat{y}_k|^2}{n}$ plotted against frequency k/T; inspect k=0,…,⌊n/2⌋ for real data. Large p_k identify dominant periodic components; period=T/k (***k>0***).

7. IMPLIED VOLATILITY & NONLINEAR EQUATIONS

Black–Scholes call and implied volatility

$\tau = T - t > 0$. $c = SN(d_1) - Ke^{-r\tau}N(d_2)$;

$d_1 = \frac{\ln(S/K)+r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}$; $N(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\xi^2/2}\,d\xi$.

Put-call parity: $c + Ke^{-r\tau} = p + S$.

Require S>0, K>0, σ>0, τ>0; model assumptions in Topic 10.

Implied volatility: $f(\sigma) = c(\sigma) - c_{\text{mkt}} = 0, \qquad f'(\sigma) = SN'(d_1)\sqrt{\tau} > 0$.

Existence and uniqueness when $\max(S - Ke^{-r\tau}, 0) \le c_{\text{mkt}} \le S$.

$f(0^+) = \max(S - Ke^{-r\tau}, 0) - c_{\text{mkt}}, \qquad f(\infty) = S - c_{\text{mkt}}$.

Taylor and finite differences

If $f \in C^{n+1}[a,b]$, $x, x+h \in [a,b]$: $f(x+h) = \sum_{k=0}^{n} \frac{h^k}{k!} f^{(k)}(x) + O(h^{n+1})$.

$f'(x) = \frac{f(x+h)-f(x)}{h} + O(h) = \frac{f(x)-f(x-h)}{h} + O(h)$ ***Require f∈C² and shifted point in [a,b].***

$f'(x) = \frac{f(x+h)-f(x-h)}{2h} + O(h^2)$ ***Require f∈C³ and x±h∈[a,b].***

$f''(x) = \frac{f(x+h)-2f(x)+f(x-h)}{h^2} + O(h^2)$ ***Require f∈C⁴ and x±h∈[a,b].***

Machine-rounding balance:
$O(h) + O(\varepsilon/h) : h \asymp \varepsilon^{1/2}; \quad O(h^2) + O(\varepsilon/h) : h \asymp \varepsilon^{1/3}; \quad O(h^2) + O(\varepsilon/h^2) : h \asymp \varepsilon^{1/4}$.

h must be representable so x±h differs from x.

Scalar iterations

$e^{(k)} = x^{(k)} - x^*, \qquad \beta = \lim_{k\to\infty} \frac{e^{(k+1)}}{(e^{(k)})^\nu}$; for $\nu=1$ require $|\beta|<1$.

Newton: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$. Quadratic locally; ***require f(x*)=0, f′ and f″ continuous near x*, f′≠0 there, initial guess sufficiently close***; asymptotic ratio $\to |f''(x^*)|/(2|f'(x^*)|)$.

Secant: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f[x^{(k-1)}, x^{(k)}]}$; order $(1+\sqrt{5})/2$ locally.

Require two distinct initial iterates, nonzero divided differences, starts sufficiently close to a simple root.

Systems / Gauss–Newton

$r: \mathbb{R}^n \to \mathbb{R}^m$: $J_{ij} = \frac{\partial r_i}{\partial x_j}, \qquad r(x+d) = r(x) + J(x)d + O(\|d\|^2)$. For m=n,

Newton–Raphson: $J(x^{(k)})d^{(k)} = -r(x^{(k)}), \qquad x^{(k+1)} = x^{(k)} + d^{(k)}$; ***J must be nonsingular at each solve; local quadratic convergence under smoothness and a sufficiently close start.***

For m≥n nonlinear least squares $\min\|r\|_2^2$: $J^T J d = -J^T r$.

Local step is uniquely defined when J has full column rank.

$J_{ij} \approx \frac{r_i(x+he_j) - r_i(x)}{h}, \qquad J_{ij} \approx \frac{r_i(x+he_j) - r_i(x-he_j)}{2h}$; use $h \approx \varepsilon^{1/2}$ forward, $h \approx \varepsilon^{1/3}$ central.

8. NUMERICAL INTEGRATION

Probability integrals and quadrature

PDF p: $p \ge 0$, $\int_{\mathbb{R}} p = 1$. $E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x)\,dx, \qquad P(x) = \int_{-\infty}^{x} p(\xi)\,d\xi$.

If P is differentiable and strictly increasing, y=P(x) gives $E[f(X)] = \int_0^1 f(P^{-1}(y))\,dy$.

$I(f) = \int_a^b f(x)\,dx, \qquad Q(f) = \sum_i w_i f(\xi_i), \qquad E = I - Q$. Degree of precision m: exact for all polynomials degree≤m and not all of degree m+1.

Composite rules

Grid $a = x_0 < \cdots < x_N = b$, $h_i = x_i - x_{i-1}$. Trapezoidal: $Q_N^{\text{trap}} = \sum_{i=1}^{N} \frac{h_i}{2}(f_{i-1} + f_i)$. Uniform $h = (b-a)/N$: $h[\frac{1}{2}f_0 + \sum_{i=1}^{N-1} f_i + \frac{1}{2}f_N]$.

Midpoint: $Q_N^{\text{mid}} = \sum_{i=1}^{N} h_i f\left(\frac{x_{i-1}+x_i}{2}\right)$.

$|E_N^{\text{trap}}| \le \frac{(b-a)Mh^2}{12}, \qquad |E_N^{\text{mid}}| \le \frac{(b-a)Mh^2}{24}$. ***Require f∈C²[a,b] for O(h²).*** Both have degree 1.

Composite Simpson, ***uniform grid and even N***:

$Q_N^S = \frac{h}{3}[f_0 + 4f_1 + 2f_2 + \cdots + 4f_{N-1} + f_N]$. Degree 3; $|E_N^S| \le \frac{(b-a)\max|f^{(4)}|h^4}{2880}$.

Require f∈C⁴[a,b] for O(h⁴).

$|E_{2N}| \approx \frac{|E_N|}{2^r}, \qquad r \approx \log_2\left(\frac{|E_N|}{|E_{2N}|}\right)$.

Gauss–Legendre

On [−1,1], N nodes/weights chosen exact through x^{2N-1}: degree $2N-1$.

$N=2:\ \xi = \pm\frac{1}{\sqrt{3}},\ w=1; \qquad N=3:\ \xi = 0, \pm\sqrt{\frac{3}{5}},\ w = \frac{8}{9}, \frac{5}{9}, \frac{5}{9}$.

$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\sum_i w_i f\left(\frac{a+b}{2} + \frac{b-a}{2}\xi_i\right)$.

Non-smooth and multidimensional

Formal orders are not guaranteed when required derivatives are missing/unbounded. Split at nonsmooth points or change variables to regularise before applying a high-order rule.

Tensor product: $\int_{[0,1]^2} f(x,y)\,dx\,dy \approx \sum_{i,j} w_i w_j f(\xi_i, \xi_j)$. A d-fold (N+1)-point rule uses $(N+1)^d$ points (**curse of dimensionality**).

9. RANDOM NUMBERS & SIMULATION — DISTRIBUTIONS

True random: physical source. Pseudorandom: deterministic, reproducible, statistically random-like. Quasirandom: deterministic, designed for multidimensional uniformity.

$X \sim U[a,b]: \quad p(x) = \frac{1}{b-a}, \quad E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$ (***a<b***); if $U \sim U[0,1]$, $Y = a + (b-a)U$.

$U \sim U[0,1], \qquad Y = F^{-1}(U) \Longrightarrow F_Y = F$. ***Use a suitable generalised inverse if strict invertibility fails.***

$Y \sim N(\mu, \sigma^2): \quad p(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}, \qquad Y = \mu + \sigma Z$ (***σ>0***). Independent sums: $E\sum X_i = \sum EX_i$; $\text{Var}\sum X_i = \sum \text{Var}X_i$. CLT: iid sums are approximately normal for large n and finite variance.

9. RANDOM NUMBERS & SIMULATION — MC/QMC

Monte Carlo

For $X \in \mathbb{R}^d$ with PDF p: $E[f] = \int f(x)p(x)\,dx$. With iid samples $x_i \sim p$, $E_N[f] = \frac{1}{N}\sum_{i=1}^{N} f(x_i)$; unbiased. RMS error $e_N[f] = \frac{\sigma[f]}{\sqrt{N}} = O(N^{-1/2})$, independent of d. $e_N[f] \le \text{tol} \quad \Longrightarrow \quad N \ge \left(\frac{\sigma[f]}{\text{tol}}\right)^2$. ***Require independent samples and finite Var[f].***

Variance reduction

Control variate: known $\mu_g = E[g]$, estimate $E[f-g] + \mu_g$; useful when $\text{Var}(f-g) \ll \text{Var}(f)$.

Antithetic pairs: ***N even***; on symmetric $\mathbb{R}^d$ pair x and −x; on $[0,1]^d$ pair x and 1−x. Odd component integrates exactly.

Importance sampling: $q(x) = g(x)p(x), \qquad E[f] \approx \frac{1}{N}\sum_{i=1}^{N} \frac{f(y_i)}{g(y_i)}, \quad y_i \sim q$.

Require q≥0, ∫q=1, and g(x)≠0 wherever f(x)≠0; finite variance of f/g under q.

Quasi-Monte Carlo

Deterministic $x_j \in [0,1]^d$: $Q_N = (1/N)\sum f(x_j)$. Koksma–Hlawka: $|\int_{[0,1]^d} f - Q_N| \le V(f)D_N^*$. ***Requires bounded variation V(f).***

$D_N^* = \sup_{y \in [0,1]^d} \left| \prod_{i=1}^{d} y_i - \frac{\#\{x_j \in [0,y)\}}{N} \right|, \qquad D_N^* \le C\frac{(\log N)^d}{N}$; C depends on d.

10. STOCHASTIC INTEGRATION & SDEs — FOUNDATIONS

Random walk / Wiener process

$t_n = n\Delta t$, $\Delta t = T/N$:

$\Delta X_n = \mu\Delta t + \sigma(\Delta t)^\gamma Z_n, \quad E[X_N - X_0] = \mu T, \quad \text{Var}(X_N - X_0) = \sigma^2 T(\Delta t)^{2\gamma-1}$;

iid $Z_n \sim N(0,1)$. Finite nonzero limit requires $\gamma = \frac{1}{2}$.

$S_{n+1} = S_n\left(1 + \mu\Delta t + \sigma\sqrt{\Delta t}\,Z_n\right)$.

Wiener W: W(0)=0; a.s. continuous paths; disjoint increments independent;

$W(t) - W(s) \sim N(0, t-s), \qquad W_{j+1} = W_j + \sqrt{\Delta t}\,Z_j$.

Itô integral / lemma

$\int_0^T f(W(t),t)\,dW(t) = \lim_{N\to\infty} \sum_{n=0}^{N-1} f(W(t_n),t_n)[W(t_{n+1}) - W(t_n)]$ (***left endpoints; adapted integrand***).

$\sum_n (\Delta W_n)^2 \longrightarrow T, \qquad \int_0^T W\,dW = \frac{1}{2}W(T)^2 - \frac{1}{2}T$.

$dX = f(X,t)\,dt + g(X,t)\,dW, \qquad X(0) = X_0$. For Y=H(X,t):

$dY = \left(H_t + \frac{1}{2}g^2 H_{xx}\right)dt + H_x\,dX$. ***Require H continuously differentiable in t and twice continuously differentiable in x.***

10. STOCHASTIC INTEGRATION & BLACK–SCHOLES

Geometric Brownian motion

$dS/S = \mu\,dt + \sigma\,dW$, ***S₀>0, σ>0***. $S(t) = S_0\exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right]$;

$\ln\left(\frac{S(t)}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right)$; S(t) is lognormal and positive.

Black–Scholes PDE and conditions

$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$, S>0, 0<t<T. Delta hedge uses $f = V_S$ short shares. ***Assumptions: GBM underlying; constant known r,σ; no dividends, transaction costs, taxes or arbitrage; continuous trading and short selling allowed.***

$c(S,T) = \max(S-K,0),\ c(0,t) = 0,\ c \sim S - Ke^{-r(T-t)}; \qquad p(S,T) = \max(K-S,0),\ p(0,t) = Ke^{-r(T-t)},\ p \to 0$.

Put-call parity: $S + p = c + Ke^{-r(T-t)}$ (***European options, same K,T***).

Log-price/time transformation

$y = \ln(S/K)$, $\tau = T - t$, $V(S,t) = U(y,\tau)$:

$-U_\tau + \left(r - \frac{1}{2}\sigma^2\right)U_y + \frac{1}{2}\sigma^2 U_{yy} = rU$; $y \in \mathbb{R}$, $\tau \in [0,T]$.

$U(y,0) = K\max(e^y - 1, 0), \quad U \to 0\ (y \to -\infty), \quad U \sim Ke^y - Ke^{-r\tau}\ (y \to \infty)$;

$U(y,0) = K\max(1 - e^y, 0), \quad U \sim Ke^{-r\tau}\ (y \to -\infty), \quad U \to 0\ (y \to \infty)$.

11. FDMs FOR BLACK–SCHOLES

Grid and differences

Truncate $S \in [0, S_{\max}]$, ***S_{max}>K and sufficiently large***.

$\Delta S = \frac{S_{\max}}{N_S + 1}, \quad S_j = j\Delta S; \qquad \Delta t = \frac{T}{N_t}, \quad t_n = n\Delta t$.

$V_S \approx \frac{V_{j+1} - V_{j-1}}{2\Delta S}, \qquad V_{SS} \approx \frac{V_{j+1} - 2V_j + V_{j-1}}{(\Delta S)^2}$; each $O((\Delta S)^2)$. Time one-sided error $O(\Delta t)$. Boundary/terminal values from Topic 10.

Backward-time explicit scheme (slides: backward Euler)

$\rho = \frac{\Delta t}{2(\Delta S)^2}, \quad \alpha_j = (\sigma^2 S_j^2 - rS_j\Delta S)\rho, \quad \beta_j = 1 - r\Delta t - 2\sigma^2 S_j^2\rho, \quad \gamma_j = (\sigma^2 S_j^2 + rS_j\Delta S)\rho$.

$V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n$. Accuracy $O(\Delta t + (\Delta S)^2)$ when stable.

Sufficient stability: $\sigma^2 \ge r, \qquad \frac{\Delta t}{(\Delta S)^2} \le \min_{1\le j\le N_S} \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2}$; then $\alpha, \beta, \gamma \ge 0$ and $|\alpha| + |\beta| + |\gamma| \le 1$. Under this mesh relation overall order is $O((\Delta S)^2)$.

Forward-time implicit scheme (slides: forward Euler)

$\alpha_j = \frac{j\Delta t}{2}(r - \sigma^2 j), \quad \beta_j = 1 + r\Delta t + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2}, \quad \gamma_j = -\frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} - \frac{rS_j\Delta t}{2\Delta S}$. Solve tridiagonal $\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1}$ with boundary terms in RHS, backwards $n = N_t - 1, \ldots, 0$. Accuracy $O(\Delta t + (\Delta S)^2)$; unconditionally stable. ***σ²≥r is sufficient for strict diagonal dominance/nonsingularity, not necessary.*** Factor once if r,σ time-independent; total $O(N_t N_S)$.

Crank–Nicolson and Greeks

At $t_{n+1/2}$: $V_j^{n+1/2} = \frac{V_j^n + V_j^{n+1}}{2}, \qquad V_t \approx \frac{V_j^{n+1} - V_j^n}{\Delta t}$; use midpoint value in all S terms. Implicit; accuracy $O\left((\Delta t)^2 + (\Delta S)^2\right)$.

$\Delta = V_S, \qquad \Gamma = V_{SS}, \qquad \Theta = V_t, \qquad \rho = V_r, \qquad \text{Vega} = V_\sigma$.